

Chapter 4 was amended by [Local Law 77 of 2023](#). This law has an effective date of June 10, 2023.

CHAPTER 4

GAS PIPING INSTALLATIONS

SECTION FGC 401 GENERAL

401.1 Scope. This chapter shall govern the design, installation, modification and maintenance of fuel-gas piping systems. The scope covered by this chapter includes piping systems from the point of delivery to the connections with the appliances and includes the design, materials, components, fabrication, assembly, installation, testing, inspection, operation and maintenance of such piping systems.

401.1.1 Meters and service piping. Service piping includes the fuel-gas piping up to the point of delivery. Meters and service piping shall comply with the requirements of Appendix E. In addition, service piping located within buildings shall be designed and installed in accordance with the structural integrity, firestopping, and fire protection provisions of the *New York City Building Code*.

401.1.2 Meters for gas consumption monitoring. Approved gas meters shall be permitted to be installed for gas consumption monitoring, energy measurement, verification and analysis. Gas meters shall be listed in accordance with ANSI B109.1, ANSI B109.2, or ANSI B109.3 and installed in accordance with the manufacturer's instructions. A full size valve bypass shall be provided around gas meters for maintenance and repair.

401.2 Reserved.

401.3 Modifications to existing systems. In modifying or adding to existing piping systems, sizes shall be maintained in accordance with this chapter.

401.4 Additional appliances. Where an additional appliance is to be served, the existing piping shall be checked to determine if it has adequate capacity for all appliances served. If inadequate, the existing system shall be enlarged as required or separate piping of adequate capacity shall be provided.

401.5 Identification. All piping installed in new construction and all new piping installed in existing buildings, whether or not the piping is intended to be enclosed when construction is completed, shall be identified by a yellow label marked "Gas" in black letters. Where the installation requires a gas test, such labeling shall be completed prior to such test. Labels shall be provided in accordance with ASME A13.1 and the marking shall be spaced at intervals not exceeding 5 feet (1524 mm). The marking shall not be required on pipe located in the same room as the appliance served.

401.6 Interconnections. Where two or more meters are installed on the same premises but supply separate consumers, the piping systems shall not be interconnected on the outlet side of the meters.

401.7 Piping meter identification. Piping from multiple meter installations shall be marked with an approved permanent identification by the installer so that the piping system supplied by each meter is readily identifiable. Such marking shall be in accordance with ASME A13.1.

401.8 Minimum sizes. Pipe utilized for the installation, extension and alteration of any piping system shall be sized to supply the full number of outlets for the intended purpose and shall be sized in accordance with Section 402.

401.9 Identification. Each length of pipe and each pipe fitting, utilized in a fuel gas system, shall bear the identification of the manufacturer.

401.10 Third-party testing and certification. Piping, tubing and fittings shall comply with the applicable referenced standards, specifications and performance criteria of this code and shall be identified in accordance with Section 401.9.

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Piping, tubing and fittings shall either be tested by an approved third-party testing agency or certified by an approved third-party certification agency.

SECTION FGC 402 PIPE SIZING

402.1 General considerations. Piping systems shall be of such size and so installed as to provide a supply of gas sufficient to meet the maximum demand and supply gas to each appliance inlet at not less than the minimum supply pressure required by the appliance.

402.2 Maximum gas demand. The volumetric flow rate of gas to be provided shall be the sum of the maximum input of the appliances served.

The total connected hourly load shall be used as the basis for pipe sizing, assuming that all appliances could be operating at full capacity simultaneously. Where a diversity of load can be established, pipe sizing shall be permitted to be based on such loads.

The volumetric flow rate of gas to be provided shall be adjusted for altitude where the installation is above 2,000 feet (610 m) in elevation.

402.3 Sizing. Gas piping shall be sized in accordance with one of the following:

1. Pipe sizing tables or sizing equations in accordance with Section 402.4.
2. The sizing tables included in a listed piping system's manufacturer's instructions.
3. Other approved engineering methods.
4. Individual outlets to gas ranges shall not be less than $\frac{3}{4}$ inches (19.1 mm) NPS.

402.4 Sizing tables and equations. Where Tables 402.4(1) through 402.4(7) are used to size piping or tubing, the pipe length shall be determined in accordance with Section 402.4.1, 402.4.2 or 402.4.3.

Where Equations 4-1 and 4-2 are used to size piping or tubing, the pipe or tubing shall have smooth inside walls and the pipe length shall be determined in accordance with Section 402.4.1, 402.4.2 or 402.4.3.

1. Low-pressure gas equation (Less than $1\frac{1}{2}$ pounds per square inch (psi) (10.3 kPa)):

$$D = \frac{Q^{0.381}}{19.17 \left(\frac{\Delta H}{C_r \times L} \right)^{0.206}} \quad \text{(Equation 4-1)}$$

2. High-pressure gas equation ($1\frac{1}{2}$ psi (10.3 kPa) and above):

$$D = \frac{Q^{0.381}}{18.93 \left\{ \frac{(P_1^2 - P_2^2) \times Y}{C_r \times L} \right\}^{0.206}} \quad \text{(Equation 4-2)}$$

where:

D = Inside diameter of pipe, inches (mm).

Q = Input rate appliance(s), cubic feet per hour at 60°F (15.6°C) and 30-inch mercury column.

P_1 = Upstream pressure, psia ($P_1 + 14.7$).

P_2 = Downstream pressure, psia ($P_2 + 14.7$).

L = Equivalent length of pipe, feet.

ΔH = Pressure drop, inch water column (27.7 inch water column = 1 psi).

TABLE 402.4
C_r AND Y VALUES FOR NATURAL GAS
AT STANDARD CONDITIONS

GAS	EQUATION FACTORS	
	C _r	Y
Natural gas	0.6094	0.9992

For SI: 1 cubic foot = 0.028 m³, 1 foot = 305 mm,

1-inch water column = 0.2488 kPa,

1 pound per square inch = 6.895 kPa,

1 British thermal unit per hour = 0.293 W.

402.4.1 Longest length method. The pipe size of each section of gas piping shall be determined using the longest length of piping from the point of delivery to the most remote outlet and the load of the section.

402.4.2 Branch length method. Pipe shall be sized as follows:

1. Pipe size of each section of the longest pipe run from the point of delivery to the most remote outlet shall be determined using the longest run of piping and the load of the section.
2. The pipe size of each section of branch piping not previously sized shall be determined using the length of piping from the point of delivery to the most remote outlet in each branch and the load of the section.

402.4.3 Hybrid pressure. The pipe size for each section of higher pressure gas piping shall be determined using the longest length of piping from the point of delivery to the most remote line pressure regulator. The pipe size from the line pressure regulator to each outlet shall be determined using the length of piping from the regulator to the most remote outlet served by the regulator.

TABLE 402.4(1)
SCHEDULE 40 METALLIC PIPE

Gas			Natural											
Inlet Pressure			Less than 2 psi											
Pressure Drop			0.3 in. w. c.											
Specific Gravity			0.60											
PIPE SIZE (inch)														
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4	5	6	8	10	12
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026	5.047	6.065	7.981	10.020	11.938
Length (ft)	Capacity in Cubic Feet of Gas Per Hour													
10	131	273	514	1,060	1,580	3,050	4,860	8,580	17,500	31,700	51,300	105,000	191,000	303,000
20	90	188	353	726	1,090	2,090	3,340	5,900	12,000	21,800	35,300	72,400	132,000	208,000
30	72	151	284	583	873	1,680	2,680	4,740	9,660	17,500	28,300	58,200	106,000	167,000
40	62	129	243	499	747	1,440	2,290	4,050	8,270	15,000	24,200	49,800	90,400	143,000
50	55	114	215	442	662	1,280	2,030	3,590	7,330	13,300	21,500	44,100	80,100	127,000
60	50	104	195	400	600	1,160	1,840	3,260	6,640	12,000	19,500	40,000	72,600	115,000
70	46	95	179	368	552	1,060	1,690	3,000	6,110	11,100	17,900	36,800	66,800	106,000
80	42	89	167	343	514	989	1,580	2,790	5,680	10,300	16,700	34,200	62,100	98,400
90	40	83	157	322	482	928	1,480	2,610	5,330	9,650	15,600	32,100	58,300	92,300
100	38	79	148	304	455	877	1,400	2,470	5,040	9,110	14,800	30,300	55,100	87,200
125	33	70	131	269	403	777	1,240	2,190	4,460	8,080	13,100	26,900	48,800	77,300
150	30	63	119	244	366	704	1,120	1,980	4,050	7,320	11,900	24,300	44,200	70,000
175	28	58	109	224	336	648	1,030	1,820	3,720	6,730	10,900	22,400	40,700	64,400
200	26	54	102	209	313	602	960	1,700	3,460	6,260	10,100	20,800	37,900	59,900
250	23	48	90	185	277	534	851	1,500	3,070	5,550	8,990	18,500	33,500	53,100
300	21	43	82	168	251	484	771	1,360	2,780	5,030	8,150	16,700	30,400	48,100
350	19	40	75	154	231	445	709	1,250	2,560	4,630	7,490	15,400	28,000	44,300

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400	18	37	70	143	215	414	660	1,170	2,380	4,310	6,970	14,300	26,000	41,200
450	17	35	66	135	202	389	619	1,090	2,230	4,040	6,540	13,400	24,400	38,600
500	16	33	62	127	191	367	585	1,030	2,110	3,820	6,180	12,700	23,100	36,500
550	15	31	59	121	181	349	556	982	2,000	3,620	5,870	12,100	21,900	34,700
600	14	30	56	115	173	333	530	937	1,910	3,460	5,600	11,500	20,900	33,100
650	14	29	54	110	165	318	508	897	1,830	3,310	5,360	11,000	20,000	31,700
700	13	27	52	106	159	306	488	862	1,760	3,180	5,150	10,600	19,200	30,400
750	13	26	50	102	153	295	470	830	1,690	3,060	4,960	10,200	18,500	29,300
800	12	26	48	99	148	285	454	802	1,640	2,960	4,790	9,840	17,900	28,300
850	12	25	46	95	143	275	439	776	1,580	2,860	4,640	9,530	17,300	27,400
900	11	24	45	93	139	267	426	752	1,530	2,780	4,500	9,240	16,800	26,600
950	11	23	44	90	135	259	413	731	1,490	2,700	4,370	8,970	16,300	25,800
1,000	11	23	43	87	131	252	402	711	1,450	2,620	4,250	8,720	15,800	25,100
1,100	10	21	40	83	124	240	382	675	1,380	2,490	4,030	8,290	15,100	23,800
1,200	NA	20	39	79	119	229	364	644	1,310	2,380	3,850	7,910	14,400	22,700
1,300	NA	20	37	76	114	219	349	617	1,260	2,280	3,680	7,570	13,700	21,800
1,400	NA	19	35	73	109	210	335	592	1,210	2,190	3,540	7,270	13,200	20,900
1,500	NA	18	34	70	105	203	323	571	1,160	2,110	3,410	7,010	12,700	20,100
1,600	NA	18	33	68	102	196	312	551	1,120	2,030	3,290	6,770	12,300	19,500
1,700	NA	17	32	66	98	189	302	533	1,090	1,970	3,190	6,550	11,900	18,800
1,800	NA	16	31	64	95	184	293	517	1,050	1,910	3,090	6,350	11,500	18,300
1,900	NA	16	30	62	93	178	284	502	1,020	1,850	3,000	6,170	11,200	17,700
2,000	NA	16	29	60	90	173	276	488	1,000	1,800	2,920	6,000	10,900	17,200

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,

1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Notes:

1. NA means a flow of less than 10 cfh.
2. All table entries have been rounded to three significant digits.

TABLE 402.4(2)
SCHEDULE 40 METALLIC PIPE

Gas				Natural										
Inlet Pressure				Less than 2 psi										
Pressure Drop				0.5 in w.c.										
Specific Gravity				0.60										
PIPE SIZE (inch)														
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4	5	6	8	10	12
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026	5.047	6.065	7.981	10.020	11.938
Length (ft)	Capacity in Cubic Feet of Gas Per Hour													
10	172	360	678	1,390	2,090	4,020	6,400	11,300	23,100	41,800	67,600	139,000	252,000	399,000
20	118	247	466	957	1,430	2,760	4,400	7,780	15,900	28,700	46,500	95,500	173,000	275,000
30	95	199	374	768	1,150	2,220	3,530	6,250	12,700	23,000	37,300	76,700	139,000	220,000
40	81	170	320	657	985	1,900	3,020	5,350	10,900	19,700	31,900	65,600	119,000	189,000
50	72	151	284	583	873	1,680	2,680	4,740	9,660	17,500	28,300	58,200	106,000	167,000
60	65	137	257	528	791	1,520	2,430	4,290	8,760	15,800	25,600	52,700	95,700	152,000
70	60	126	237	486	728	1,400	2,230	3,950	8,050	14,600	23,600	48,500	88,100	139,000
80	56	117	220	452	677	1,300	2,080	3,670	7,490	13,600	22,000	45,100	81,900	130,000
90	52	110	207	424	635	1,220	1,950	3,450	7,030	12,700	20,600	42,300	76,900	122,000
100	50	104	195	400	600	1,160	1,840	3,260	6,640	12,000	19,500	40,000	72,600	115,000
125	44	92	173	355	532	1,020	1,630	2,890	5,890	10,600	17,200	35,400	64,300	102,000
150	40	83	157	322	482	928	1,480	2,610	5,330	9,650	15,600	32,100	58,300	92,300
175	37	77	144	296	443	854	1,360	2,410	4,910	8,880	14,400	29,500	53,600	84,900

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200	34	71	134	275	412	794	1,270	2,240	4,560	8,260	13,400	27,500	49,900	79,000
250	30	63	119	244	366	704	1,120	1,980	4,050	7,320	11,900	24,300	44,200	70,000
300	27	57	108	221	331	638	1,020	1,800	3,670	6,630	10,700	22,100	40,100	63,400
350	25	53	99	203	305	587	935	1,650	3,370	6,100	9,880	20,300	36,900	58,400
400	23	49	92	189	283	546	870	1,540	3,140	5,680	9,190	18,900	34,300	54,300
450	22	46	86	177	266	512	816	1,440	2,940	5,330	8,620	17,700	32,200	50,900
500	21	43	82	168	251	484	771	1,360	2,780	5,030	8,150	16,700	30,400	48,100
550	20	41	78	159	239	459	732	1,290	2,640	4,780	7,740	15,900	28,900	45,700
600	19	39	74	152	228	438	699	1,240	2,520	4,560	7,380	15,200	27,500	43,600
650	18	38	71	145	218	420	669	1,180	2,410	4,360	7,070	14,500	26,400	41,800
700	17	36	68	140	209	403	643	1,140	2,320	4,190	6,790	14,000	25,300	40,100
750	17	35	66	135	202	389	619	1,090	2,230	4,040	6,540	13,400	24,400	38,600
800	16	34	63	130	195	375	598	1,060	2,160	3,900	6,320	13,000	23,600	37,300
850	16	33	61	126	189	363	579	1,020	2,090	3,780	6,110	12,600	22,800	36,100
900	15	32	59	122	183	352	561	992	2,020	3,660	5,930	12,200	22,100	35,000
950	15	31	58	118	178	342	545	963	1,960	3,550	5,760	11,800	21,500	34,000
1,000	14	30	56	115	173	333	530	937	1,910	3,460	5,600	11,500	20,900	33,100
1,100	14	28	53	109	164	316	503	890	1,810	3,280	5,320	10,900	19,800	31,400
1,200	13	27	51	104	156	301	480	849	1,730	3,130	5,070	10,400	18,900	30,000
1,300	12	26	49	100	150	289	460	813	1,660	3,000	4,860	9,980	18,100	28,700
1,400	12	25	47	96	144	277	442	781	1,590	2,880	4,670	9,590	17,400	27,600
1,500	11	24	45	93	139	267	426	752	1,530	2,780	4,500	9,240	16,800	26,600
1,600	11	23	44	89	134	258	411	727	1,480	2,680	4,340	8,920	16,200	25,600
1,700	11	22	42	86	130	250	398	703	1,430	2,590	4,200	8,630	15,700	24,800
1,800	10	22	41	84	126	242	386	682	1,390	2,520	4,070	8,370	15,200	24,100
1,900	10	21	40	81	122	235	375	662	1,350	2,440	3,960	8,130	14,800	23,400
2,000	NA	20	39	79	119	229	364	644	1,310	2,380	3,850	7,910	14,400	22,700

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,
1 British thermal unit per hour = 0.293 1 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Notes:

1. NA means a flow of less than 10 cfh.
2. All table entries have been rounded to three significant digits.

**TABLE 402.4(3)
SCHEDULE 40 METALLIC PIPE**

Gas			Natural						
Inlet Pressure			Less than 2.0 psi						
Pressure Drop			3.0 in. w.c.						
Specific Gravity			0.60						
INTENDED USE: Initial supply pressure of 8.0 inches w.c. or greater									
PIPE SIZE (inch)									
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026
Length (ft)	Capacity in Cubic Feet of Gas Per Hour								
10	454	949	1,790	3,670	5,500	10,600	16,900	29,800	60,800
20	312	652	1,230	2,520	3,780	7,280	11,600	20,500	41,800
30	250	524	986	2,030	3,030	5,840	9,310	16,500	33,600
40	214	448	844	1,730	2,600	5,000	7,970	14,100	28,700
50	190	397	748	1,540	2,300	4,430	7,070	12,500	25,500
60	172	360	678	1,390	2,090	4,020	6,400	11,300	23,100
70	158	331	624	1,280	1,920	3,700	5,890	10,400	21,200
80	147	308	580	1,190	1,790	3,440	5,480	9,690	19,800
90	138	289	544	1,120	1,680	3,230	5,140	9,090	18,500

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100	131	273	514	1,060	1,580	3,050	4,860	8,580	17,500
125	116	242	456	936	1,400	2,700	4,300	7,610	15,500
150	105	219	413	848	1,270	2,450	3,900	6,890	14,100
175	96	202	380	780	1,170	2,250	3,590	6,340	12,900
200	90	188	353	726	1,090	2,090	3,340	5,900	12,000
250	80	166	313	643	964	1,860	2,960	5,230	10,700
300	72	151	284	583	873	1,680	2,680	4,740	9,700
350	66	139	261	536	803	1,550	2,470	4,360	8,900
400	62	129	243	499	747	1,440	2,290	4,060	8,300
450	58	121	228	468	701	1,350	2,150	3,800	7,800
500	55	114	215	442	662	1,280	2,030	3,590	7,300
550	52	109	204	420	629	1,210	1,930	3,410	7,000
600	50	104	195	400	600	1,160	1,840	3,260	6,640
650	47	99	187	384	575	1,110	1,760	3,120	6,360
700	46	95	179	368	552	1,060	1,700	3,000	6,110
750	44	92	173	355	532	1,020	1,630	2,890	5,890
800	42	89	167	343	514	989	1,580	2,790	5,680
850	41	86	162	332	497	957	1,530	2,700	5,500
900	40	83	157	322	482	928	1,480	2,620	5,330
950	39	81	152	312	468	901	1,440	2,540	5,180
1,000	38	79	148	304	455	877	1,400	2,470	5,040
1,100	36	75	141	289	432	833	1,330	2,350	4,780
1,200	34	71	134	275	412	794	1,270	2,240	4,560
1,300	33	68	128	264	395	761	1,210	2,140	4,370
1,400	31	65	123	253	379	731	1,170	2,060	4,200
1,500	30	63	119	244	366	704	1,120	1,980	4,050
1,600	29	61	115	236	353	680	1,080	1,920	3,910
1,700	28	59	111	228	342	658	1,050	1,850	3,780
1,800	27	57	108	221	331	638	1,020	1,800	3,670
1,900	27	56	105	215	322	619	987	1,750	3,560
2,000	26	54	102	209	313	602	960	1,700	3,460

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,
1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Note: All table entries have been rounded to three significant digits.

TABLE 402.4(4)
SCHEDULE 40 METALLIC PIPE

Gas			Natural						
Inlet Pressure			Less than 2.0 psi						
Pressure Drop			6.0 in. w.c.						
Specific Gravity			0.60						
INTENDED USE: Initial supply pressure of 11.0 inches w.c. or greater									
PIPE SIZE (inch)									
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026
Length (ft)	Capacity in Cubic Feet of Gas Per Hour								
10	660	1,380	2,600	5,340	8,000	15,400	24,600	43,400	88,500
20	454	949	1,790	3,670	5,500	10,600	16,900	29,900	60,800
30	364	762	1,440	2,950	4,420	8,500	13,600	24,000	48,900
40	312	652	1,230	2,520	3,780	7,280	11,600	20,500	41,800
50	276	578	1,090	2,240	3,350	6,450	10,300	18,200	37,100
60	250	524	986	2,030	3,030	5,840	9,310	16,500	33,600
70	230	482	907	1,860	2,790	5,380	8,570	15,100	30,900

GAS PIPING INSTALLATIONS

80	214	448	844	1,730	2,600	5,000	7,970	14,100	28,700
90	201	420	792	1,630	2,440	4,690	7,480	13,200	27,000
100	190	397	748	1,540	2,300	4,430	7,070	12,500	25,500
125	168	352	663	1,360	2,040	3,930	6,260	11,100	22,600
150	153	319	601	1,230	1,850	3,560	5,670	10,000	20,500
175	140	293	553	1,140	1,700	3,280	5,220	9,230	18,800
200	131	273	514	1,060	1,580	3,050	4,860	8,580	17,500
250	116	242	456	936	1,400	2,700	4,300	7,610	15,500
300	105	219	413	848	1,270	2,450	3,900	6,890	14,100
350	96	202	380	780	1,170	2,250	3,590	6,340	12,900
400	90	188	353	726	1,090	2,090	3,340	5,900	12,000
450	84	176	332	681	1,020	1,970	3,130	5,540	11,300
500	80	166	313	643	964	1,860	2,960	5,230	10,700
550	76	158	297	611	915	1,760	2,810	4,970	10,100
600	72	151	284	583	873	1,680	2,680	4,740	9,660
650	69	144	272	558	836	1,610	2,570	4,540	9,250
700	66	139	261	536	803	1,550	2,470	4,360	8,890
750	64	134	252	516	774	1,490	2,380	4,200	8,560
800	62	129	243	499	747	1,440	2,290	4,060	8,270
850	60	125	235	483	723	1,390	2,220	3,920	8,000
900	58	121	228	468	701	1,350	2,150	3,800	7,760
950	56	118	221	454	681	1,310	2,090	3,700	7,540
1,000	55	114	215	442	662	1,280	2,030	3,590	7,330
1,100	52	109	204	420	629	1,210	1,930	3,410	6,960
1,200	50	104	195	400	600	1,160	1,840	3,260	6,640
1,300	47	99	187	384	575	1,100	1,760	3,120	6,360
1,400	46	95	179	368	552	1,060	1,700	3,000	6,110
1,500	44	92	173	355	532	1,020	1,630	2,890	5,890
1,600	42	89	167	343	514	989	1,580	2,790	5,680
1,700	41	86	162	332	497	957	1,530	2,700	5,500
1,800	40	83	157	322	482	928	1,480	2,620	5,330
1,900	39	81	152	312	468	901	1,440	2,540	5,180
2,000	38	79	148	304	455	877	1,400	2,470	5,040

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,
1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Note: All table entries have been rounded to three significant digits.

TABLE 402.4(5)
SCHEDULE 40 METALLIC PIPE

Gas			Natural						
Inlet Pressure			2.0 psi						
Pressure Drop			1.0 psi						
Specific Gravity			0.60						
PIPE SIZE (inch)									
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026
Length (ft)	Capacity in Cubic Feet of Gas Per Hour								
10	1,510	3,040	5,560	11,400	17,100	32,900	52,500	92,800	189,000
20	1,070	2,150	3,930	8,070	12,100	23,300	37,100	65,600	134,000
30	869	1,760	3,210	6,590	9,880	19,000	30,300	53,600	109,000
40	753	1,520	2,780	5,710	8,550	16,500	26,300	46,400	94,700
50	673	1,360	2,490	5,110	7,650	14,700	23,500	41,500	84,700
60	615	1,240	2,270	4,660	6,980	13,500	21,400	37,900	77,300

GAS PIPING INSTALLATIONS

70	569	1,150	2,100	4,320	6,470	12,500	19,900	35,100	71,600
80	532	1,080	1,970	4,040	6,050	11,700	18,600	32,800	67,000
90	502	1,010	1,850	3,810	5,700	11,000	17,500	30,900	63,100
100	462	934	1,710	3,510	5,260	10,100	16,100	28,500	58,200
125	414	836	1,530	3,140	4,700	9,060	14,400	25,500	52,100
150	372	751	1,370	2,820	4,220	8,130	13,000	22,900	46,700
175	344	695	1,270	2,601	3,910	7,530	12,000	21,200	43,300
200	318	642	1,170	2,410	3,610	6,960	11,100	19,600	40,000
250	279	583	1,040	2,140	3,210	6,180	9,850	17,400	35,500
300	253	528	945	1,940	2,910	5,600	8,920	15,800	32,200
350	232	486	869	1,790	2,670	5,150	8,210	14,500	29,600
400	216	452	809	1,660	2,490	4,790	7,640	13,500	27,500
450	203	424	759	1,560	2,330	4,500	7,170	12,700	25,800
500	192	401	717	1,470	2,210	4,250	6,770	12,000	24,400
550	182	381	681	1,400	2,090	4,030	6,430	11,400	23,200
600	174	363	650	1,330	2,000	3,850	6,130	10,800	22,100
650	166	348	622	1,280	1,910	3,680	5,870	10,400	21,200
700	160	334	598	1,230	1,840	3,540	5,640	9,970	20,300
750	154	322	576	1,180	1,770	3,410	5,440	9,610	19,600
800	149	311	556	1,140	1,710	3,290	5,250	9,280	18,900
850	144	301	538	1,100	1,650	3,190	5,080	8,980	18,300
900	139	292	522	1,070	1,600	3,090	4,930	8,710	17,800
950	135	283	507	1,040	1,560	3,000	4,780	8,460	17,200
1,000	132	275	493	1,010	1,520	2,920	4,650	8,220	16,800
1,100	125	262	468	960	1,440	2,770	4,420	7,810	15,900
1,200	119	250	446	917	1,370	2,640	4,220	7,450	15,200
1,300	114	239	427	878	1,320	2,530	4,040	7,140	14,600
1,400	110	230	411	843	1,260	2,430	3,880	6,860	14,000
1,500	106	221	396	812	1,220	2,340	3,740	6,600	13,500
1,600	102	214	382	784	1,180	2,260	3,610	6,380	13,000
1,700	99	207	370	759	1,140	2,190	3,490	6,170	12,600
1,800	96	200	358	736	1,100	2,120	3,390	5,980	12,200
1,900	93	195	348	715	1,070	2,060	3,290	5,810	11,900
2,000	91	189	339	695	1,040	2,010	3,200	5,650	11,500

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,

1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Note: All table entries have been rounded to three significant digits.

**TABLE 402.4(6)
SCHEDULE 40 METALLIC PIPE**

Gas			Natural						
Inlet Pressure			3.0 psi						
Pressure Drop			2.0 psi						
Specific Gravity			0.60						
PIPE SIZE (inch)									
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026
Length (ft)	Capacity in Cubic Feet of Gas Per Hour								
10	2,350	4,920	9,270	19,000	28,500	54,900	87,500	155,000	316,000
20	1,620	3,380	6,370	13,100	19,600	37,700	60,100	106,000	217,000
30	1,300	2,720	5,110	10,500	15,700	30,300	48,300	85,400	174,000
40	1,110	2,320	4,380	8,990	13,500	25,900	41,300	73,100	149,000
50	985	2,060	3,880	7,970	11,900	23,000	36,600	64,800	132,000

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60	892	1,870	3,520	7,220	10,800	20,800	33,200	58,700	120,000
70	821	1,720	3,230	6,640	9,950	19,200	30,500	54,000	110,000
80	764	1,600	3,010	6,180	9,260	17,800	28,400	50,200	102,000
90	717	1,500	2,820	5,800	8,680	16,700	26,700	47,100	96,100
100	677	1,420	2,670	5,470	8,200	15,800	25,200	44,500	90,800
125	600	1,250	2,360	4,850	7,270	14,000	22,300	39,500	80,500
150	544	1,140	2,140	4,400	6,590	12,700	20,200	35,700	72,900
175	500	1,050	1,970	4,040	6,060	11,700	18,600	32,900	67,100
200	465	973	1,830	3,760	5,640	10,900	17,300	30,600	62,400
250	412	862	1,620	3,330	5,000	9,620	15,300	27,100	55,300
300	374	781	1,470	3,020	4,530	8,720	13,900	24,600	50,100
350	344	719	1,350	2,780	4,170	8,020	12,800	22,600	46,100
400	320	669	1,260	2,590	3,870	7,460	11,900	21,000	42,900
450	300	627	1,180	2,430	3,640	7,000	11,200	19,700	40,200
500	283	593	1,120	2,290	3,430	6,610	10,500	18,600	38,000
550	269	563	1,060	2,180	3,260	6,280	10,000	17,700	36,100
600	257	537	1,010	2,080	3,110	5,990	9,550	16,900	34,400
650	246	514	969	1,990	2,980	5,740	9,150	16,200	33,000
700	236	494	931	1,910	2,860	5,510	8,790	15,500	31,700
750	228	476	897	1,840	2,760	5,310	8,470	15,000	30,500
800	220	460	866	1,780	2,660	5,130	8,180	14,500	29,500
850	213	445	838	1,720	2,580	4,960	7,910	14,000	28,500
900	206	431	812	1,670	2,500	4,810	7,670	13,600	27,700
950	200	419	789	1,620	2,430	4,670	7,450	13,200	26,900
1,000	195	407	767	1,580	2,360	4,550	7,240	12,800	26,100
1,100	185	387	729	1,500	2,240	4,320	6,890	12,200	24,800
1,200	177	369	695	1,430	2,140	4,120	6,570	11,600	23,700
1,300	169	353	666	1,370	2,050	3,940	6,290	11,100	22,700
1,400	162	340	640	1,310	1,970	3,790	6,040	10,700	21,800
1,500	156	327	616	1,270	1,900	3,650	5,820	10,300	21,000
1,600	151	316	595	1,220	1,830	3,530	5,620	10,000	20,300
1,700	146	306	576	1,180	1,770	3,410	5,440	9,610	19,600
1,800	142	296	558	1,150	1,720	3,310	5,270	9,320	19,000
1,900	138	288	542	1,110	1,670	3,210	5,120	9,050	18,400
2,000	134	280	527	1,080	1,620	3,120	4,980	8,800	18,000

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,

1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Note: All table entries have been rounded to three significant digits.

TABLE 402.4(7)
SCHEDULE 40 METALLIC PIPE

Gas		Natural							
Inlet Pressure		5.0 psi							
Pressure Drop		3.5 psi							
Specific Gravity		0.60							
PIPE SIZE (inch)									
Nominal	1/2	3/4	1	1 1/4	1 1/2	2	2 1/2	3	4
Actual ID	0.622	0.824	1.049	1.380	1.610	2.067	2.469	3.068	4.026
Length (ft)	Capacity in Cubic Feet of Gas Per Hour								
10	3,190	6,430	11,800	24,200	36,200	69,700	111,000	196,000	401,000
20	2,250	4,550	8,320	17,100	25,600	49,300	78,600	139,000	283,000
30	1,840	3,720	6,790	14,000	20,900	40,300	64,200	113,000	231,000
40	1,590	3,220	5,880	12,100	18,100	34,900	55,600	98,200	200,000

GAS PIPING INSTALLATIONS

50	1,430	2,880	5,260	10,800	16,200	31,200	49,700	87,900	179,000
60	1,300	2,630	4,800	9,860	14,800	28,500	45,400	80,200	164,000
70	1,200	2,430	4,450	9,130	13,700	26,400	42,000	74,300	151,000
80	1,150	2,330	4,260	8,540	12,800	24,700	39,300	69,500	142,000
90	1,060	2,150	3,920	8,050	12,100	23,200	37,000	65,500	134,000
100	979	1,980	3,620	7,430	11,100	21,400	34,200	60,400	123,000
125	876	1,770	3,240	6,640	9,950	19,200	30,600	54,000	110,000
150	786	1,590	2,910	5,960	8,940	17,200	27,400	48,500	98,900
175	728	1,470	2,690	5,520	8,270	15,900	25,400	44,900	91,600
200	673	1,360	2,490	5,100	7,650	14,700	23,500	41,500	84,700
250	558	1,170	2,200	4,510	6,760	13,000	20,800	36,700	74,900
300	506	1,060	1,990	4,090	6,130	11,800	18,800	33,300	67,800
350	465	973	1,830	3,760	5,640	10,900	17,300	30,600	62,400
400	433	905	1,710	3,500	5,250	10,100	16,100	28,500	58,100
450	406	849	1,600	3,290	4,920	9,480	15,100	26,700	54,500
500	384	802	1,510	3,100	4,650	8,950	14,300	25,200	51,500
550	364	762	1,440	2,950	4,420	8,500	13,600	24,000	48,900
600	348	727	1,370	2,810	4,210	8,110	12,900	22,900	46,600
650	333	696	1,310	2,690	4,030	7,770	12,400	21,900	44,600
700	320	669	1,260	2,590	3,880	7,460	11,900	21,000	42,900
750	308	644	1,210	2,490	3,730	7,190	11,500	20,300	41,300
800	298	622	1,170	2,410	3,610	6,940	11,100	19,600	39,900
850	288	602	1,130	2,330	3,490	6,720	10,700	18,900	38,600
900	279	584	1,100	2,260	3,380	6,520	10,400	18,400	37,400
950	271	567	1,070	2,190	3,290	6,330	10,100	17,800	36,400
1,000	264	551	1,040	2,130	3,200	6,150	9,810	17,300	35,400
1,100	250	524	987	2,030	3,030	5,840	9,320	16,500	33,600
1,200	239	500	941	1,930	2,900	5,580	8,890	15,700	32,000
1,300	229	478	901	1,850	2,770	5,340	8,510	15,000	30,700
1,400	220	460	866	1,780	2,660	5,130	8,180	14,500	29,500
1,500	212	443	834	1,710	2,570	4,940	7,880	13,900	28,400
1,600	205	428	806	1,650	2,480	4,770	7,610	13,400	27,400
1,700	198	414	780	1,600	2,400	4,620	7,360	13,000	26,500
1,800	192	401	756	1,550	2,330	4,480	7,140	12,600	25,700
1,900	186	390	734	1,510	2,260	4,350	6,930	12,300	25,000
2,000	181	379	714	1,470	2,200	4,230	6,740	11,900	24,300

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square inch = 6.895 kPa, 1-inch water column = 0.2488 kPa,

1 British thermal unit per hour = 0.2931 W, 1 cubic foot per hour = 0.0283 m³/h, 1 degree = 0.01745 rad.

Note: All table entries have been rounded to three significant digits.

402.5 Allowable pressure drop. The design pressure loss in any piping system under maximum probable flow conditions, from the point of delivery to the inlet connection of the appliance, shall be such that the supply pressure at the appliance is greater than or equal to the minimum pressure required by the appliance.

402.6 Gas distribution pressures. The maximum design operating pressure for piping systems located inside buildings shall not exceed $\frac{1}{2}$ pounds per square inch gauge (psig) (3.5 kPa gauge).

Exceptions:

1. Pressure not exceeding 5 psig (34.5 kPa gauge) is permitted where a system supplies one or more appliances that require gas pressure in excess of $\frac{1}{2}$ psig (3.5 kPa gauge).
2. Gas pressure not exceeding 15 psig (103.4 kPa gauge) is permitted for appliances in excess of 100,000 cubic feet per hour (2830 m³/h) provided the gas distribution piping is installed as provided for in Section 404.

3. The use of pressure in excess of 15 psig (103.4 kPa gauge) shall be permitted for distribution piping provided all of the requirements of Section 406 and Appendix G are met.

SECTION FGC 403 PIPING MATERIALS

403.1 General. Materials used for piping systems shall be new and comply with the requirements of this chapter or shall be approved.

403.1.1 Pipe size and pressure limitations.

1. All requirements for installation of gas distribution piping shall be in accordance with Chapter 4. Gas distribution piping with operating pressures of 15 psig (103.4 kPa gauge) or more shall be in accordance with Appendix G.
2. Gas distribution piping operating at a pressure of over $\frac{1}{2}$ psig (3.5 kPa gauge) shall be welded.

Exception: Manufactured and listed gas trains provided with the appliance may be threaded.

3. All welding of gas distribution piping shall be subject to radiography or special inspection as set forth in Section 406.
4. Threaded piping may be used up to 4 inches (101.6 mm) in diameter at pressure no greater than $\frac{1}{2}$ psig (3.5 kPa gauge). All piping greater than 4 inches in diameter shall be welded, regardless of operating pressure.
5. All welded piping $2\frac{1}{2}$ inches (63.5 mm) or greater in diameter shall be butt-welded, and welded piping less than $2\frac{1}{2}$ inches (63.5 mm) in diameter may be socket-welded or butt-welded.

403.2 Used materials. Used pipe, fittings, valves and other materials shall not be reused.

403.3 Other materials. Material not covered by the standards specifications listed herein shall be investigated and tested to determine that it is safe and suitable for the proposed service, and, in addition, shall be recommended for that service by the manufacturer subject to approval by the commissioner.

403.4 Metallic pipe. Metallic pipe shall comply with Sections 403.4.1 through 403.4.4.

403.4.1 Cast iron. Cast-iron pipe shall not be used.

403.4.2 Steel. Carbon steel and wrought-iron pipe shall be not less than standard weight and shall comply with one of the following standards:

1. ASME B36.10M.
2. ASTM A 53/A 53M.
3. ASTM A106/A106M.

403.4.3 Copper and brass. Copper and brass pipe shall not be used.

403.4.4 Aluminum. Aluminum-alloy pipe shall not be used.

403.5 Metallic tubing. Metallic tubing shall not be used except as provided in Section 404.2.

403.5.1 Standards. Stainless steel flexible multiple leg hose assemblies shall be listed and labeled in accordance with UL 536 and designed in accordance with the manufacturer's recommendation.

403.5.2 Seismic requirements. Stainless steel flexible multiple leg hose assemblies shall be designed to withstand seismic force and displacement in accordance with Section 1613 of the *New York City Building Code*.

403.5.3 Special inspection required. The installation of stainless steel flexible multiple leg hose assemblies shall be subject to special inspection in accordance with Section 1705.12.3 of the *New York City Building Code* and Section 406 of this code.

403.6 Reserved.

GAS PIPING INSTALLATIONS

403.7 Workmanship and defects. Pipe and fittings shall be clear and free from cutting burrs and defects in structure or threading, and shall be thoroughly brushed, and chip and scale blown.

Defects in pipe and fittings shall not be repaired. Defective pipe and fittings shall be replaced (see Section 406.1.2).

403.8 Protective coating. Where in contact with material or atmosphere exerting a corrosive action, metallic piping and fittings coated with a corrosion-resistant material shall be used. External coatings or linings used on piping or components shall not be considered as adding strength.

403.9 Metallic pipe threads. Metallic pipe and fitting threads shall be taper pipe threads and shall comply with ASME B1.20.1.

403.9.1 Damaged threads. Pipe with threads that are stripped, chipped, corroded or otherwise damaged shall not be used. Where a weld opens during the operation of cutting or threading, that portion of the pipe shall not be used.

403.9.2 Number of threads. Field threading of metallic pipe shall be in accordance with Table 403.9.2.

**TABLE 403.9.2
SPECIFICATIONS FOR THREADING METALLIC PIPE**

IRON PIPE SIZE (inches)	APPROXIMATE LENGTH OF THREADED PORTION (inches)	APPROXIMATE NUMBER OF THREADS TO BE CUT
$\frac{1}{2}$	$\frac{3}{4}$	10
$\frac{3}{4}$	$\frac{3}{4}$	10
1	$\frac{7}{8}$	10
$1\frac{1}{4}$	1	11
$1\frac{1}{2}$	1	11
2	1	11
$2\frac{1}{2}$	$1\frac{1}{2}$	12
3	$1\frac{1}{2}$	12
4	$1\frac{5}{8}$	13

For SI: 1 inch = 25.4 mm

403.9.3 Thread joint compounds. Thread joint compounds shall be in non-hardening paste or tape form and shall be resistant to the action of liquefied petroleum gas or to any other chemical constituents of the gases to be conducted through the piping. Use of cotton thread (lamp wick) is prohibited. Threaded joint compound in tape form is prohibited for use at pressure regulator connections.

403.10 Metallic piping joints and fittings. The type of piping joint used shall be suitable for the pressure-temperature conditions and shall be selected giving consideration to joint tightness and mechanical strength under the service conditions. The joint shall be able to sustain the maximum end force caused by the internal pressure and any additional forces caused by temperature expansion or contraction, vibration, fatigue or the weight of the pipe and its contents.

403.10.1 Pipe joints. Pipe joints shall be threaded, flanged, or welded.

403.10.1.1 Prohibited joints and fittings. Unions and threaded close nipples shall be prohibited.

Exception: Unions shall be permitted to be installed downstream of the appliance valve in accordance with Section 411.1.6.

403.10.2 Tubing joints. Tubing joints shall not be used.

403.10.3 Flared joints. Flared joints shall not be used.

403.10.4 Metallic fittings. Metallic fittings shall comply with the following:

1. Threaded fittings in sizes larger than 4 inches (101.6 mm) shall not be used.
2. Fittings used with steel or wrought-iron pipe shall be steel or malleable iron.

3. Bushings shall not be used.

403.11 Reserved.

403.12 Flanges. Flanges and flange gaskets shall comply with Sections 403.12.1 through 403.12.7.

403.12.1 Cast-iron. Cast-iron flanges shall not be used.

403.12.2 Steel. Steel flanges shall be in accordance with ASME B16.5 or ASME B16.47.

403.12.3 Nonferrous. Nonferrous flanges shall not be used.

403.12.4 Ductile iron. Ductile-iron flanges shall be in accordance with ASME B16.42.

403.12.5 Raised face. Raised face flanges shall not be joined to flat faced cast-iron, ductile-iron or nonferrous material flanges.

403.12.6 Flange facings. Standard facings shall be permitted for use under this code. Where 150 pound (1034 kPa) pressure-rated steel flanges are bolted to Class 125 cast-iron flanges, the raised face on the steel flange shall be removed.

403.12.7 Lapped flanges. Lapped flanges shall be used only above ground or in exposed locations accessible for inspection.

403.13 Flange gaskets. Material for gaskets shall be capable of withstanding the design temperature and pressure of the piping system, and the chemical constituents of the gas being conducted, without change to its chemical and physical properties. The effects of fire exposure to the joint shall be considered in choosing material. Acceptable materials include metal (plain or corrugated), non-asbestos fiber, and spiral wound metal gaskets. Where a flanged joint is opened, the gasket shall be replaced. Full-face flange gaskets shall be used with all non-steel flanges. Rubber-faced phenolic and elastomeric gaskets shall be prohibited.

403.13.1 Metallic gaskets. Metallic flange gaskets shall be in accordance with ASME B16.20.

403.13.2 Nonmetallic gaskets. Nonmetallic flange gaskets shall be in accordance with ASME B16.21.

403.14 Flange bolts. Flange bolts and associated hardware shall be manufactured from corrosion-resistant materials.

SECTION FGC 404 PIPING SYSTEM INSTALLATION

404.1 Installation of materials. Materials used shall be installed in strict accordance with the standards under which the materials are accepted and approved. In the absence of such installation procedures, the manufacturer's instructions shall be followed. Where the requirements of referenced standards or manufacturer's instructions do not conform to minimum provisions of this code, the provisions of this code shall apply.

404.2 Pipe movement. Flammable or combustible gas piping systems shall be designed to account for pipe movement resulting from thermal changes or seismic forces.

404.2.1 Seismic requirements. Gas piping shall be designed and installed to withstand seismic forces in accordance with Section 1613 of the *New York City Building Code*.

404.2.1.1 Stainless steel flexible multiple leg hose assemblies. Stainless steel flexible multiple leg hose assemblies may be installed where pipe movement resulting from seismic forces may occur. Such assemblies shall be listed and labeled in accordance with UL 536 and shall be installed in accordance with the manufacturer's instructions.

404.2.1.2 Inspection. The installation of stainless steel flexible multiple leg hose assemblies shall be subject to special inspections in accordance with Chapter 17 of the *New York City Building Code*.

404.3 Prohibited locations. Piping shall not be installed in or through a ducted supply, return or exhaust duct, or a trash or clothes chute, chimney or gas vent, ventilating duct, dumbwaiter or elevator shaft. Piping installed downstream of the point of delivery shall not extend through any townhouse unit other than the unit served by such piping. Piping,

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fixtures, or equipment shall be located so as not to interfere with the normal operation of windows or doors and other exit openings. The following installation limitations shall apply:

1. **Stair enclosures.** Gas piping shall not be installed within a stair enclosure or required exit or exit way.
2. **Fire standpipe riser.** Gas piping shall not be installed in any shaft containing standpipe risers.
3. **Fire pump and fire pump rooms.** Gas piping, gas consumption devices, appliances or any other gas equipment shall not be installed within any space housing a fire pump. Access to gas meter rooms shall not be permitted through rooms housing a fire pump.
4. **Fire-rated construction.** Gas piping shall not be installed within a wall or assembly required to have a fire-resistance rating. Through penetrations of fire-resistance-rated assemblies shall be in accordance with Chapter 7 of the *New York City Building Code*.
5. **Public corridor.** Gas piping shall not be installed in public corridors and exit enclosures.

Exceptions:

1. Gas piping may be installed in public corridors or exit enclosures where separated by a fire-resistance-rated assembly meeting the hour rating and, if applicable, the impact-resistance rating required for the corridor or exit enclosure. Such assembly shall be rated for exposure to fire from both sides.
2. In residential buildings that do not have floors below grade, or in multi-use buildings that have a residential occupancy, gas piping may be installed in public corridors in accordance with the following:
 - 2.1. Gas piping shall be permitted to be installed within a public corridor at the lowest level of the building or the lowest residential level of the building.
 - 2.2. All gas valves located within the public corridor shall be accessible for maintenance and inspection.
 - 2.3. Gas pressure within the public corridor piping shall not exceed $\frac{1}{2}$ psi (14 inch w.c.).
 - 2.4. The public corridor shall be ventilated in accordance with the *New York City Mechanical Code*. The pipe shall not be installed in a return air plenum.
 - 2.5. Pipes must be welded.

404.4 Piping in solid partitions and walls. Concealed piping shall not be located in solid partitions and solid walls, unless installed in a ventilated chase or casing.

404.5 Fittings in concealed locations. Fittings installed in concealed locations shall be limited to the following types:

1. Threaded elbows, tees and couplings.
2. Welded fittings.

404.6 Piping through foundation wall. Underground piping, where installed below grade through the outer foundation or basement wall of a building, shall be encased in a protective pipe sleeve. Such sleeve shall extend a minimum of 1 inch (25.4 mm) beyond the inside of the foundation wall and a minimum of 4 inches (101.6 mm) beyond the outside of the foundation wall, or as otherwise required by the gas provider. The annular space between the gas piping and the sleeve shall be waterproofed and sealed.

404.7 Branch take-off. Branches shall be taken off the riser with not less than a two-elbow swing.

404.8 Piping in solid floors. Piping in solid floors shall be laid in channels in the floor and covered in a manner that will allow access to the piping with a minimum amount of damage to the building. Where such piping is subject to exposure to excessive moisture or corrosive substances, the piping shall be protected in an approved manner. As an alternative to installation in channels, the piping shall be installed in a conduit of Schedule 40 steel or wrought iron pipe installed in accordance with Section 404.8.1 or 404.8.2.

404.8.1 Conduit with one end terminating outdoors. The conduit shall extend into an occupiable portion of the building and, at the point where the conduit terminates in the building, the space between the conduit and the gas piping shall be sealed to prevent the possible entrance of any gas leakage. The conduit shall extend not less than 2 inches (50.8 mm) beyond the point where the pipe emerges from the floor. If the end sealing is capable of withstanding the full pressure of the gas pipe, the conduit shall be designed for the same pressure as the pipe. Such

conduit shall extend not less than 4 inches (101.6 mm) outside the building, shall be vented above grade to the outdoors and shall be installed so as to prevent the entrance of water and insects.

404.8.2 Conduit with both ends terminating indoors. Where the conduit originates and terminates within the same building, the conduit shall originate and terminate in an accessible portion of the building and shall not be sealed. The conduit shall extend not less than 2 inches (50.8 mm) beyond the point where the pipe emerges from the floor.

404.9 Above-ground outdoor piping. Piping installed outdoors shall be elevated not less than 3½ inches (88.9 mm) above ground and where installed across roof surfaces, shall be elevated not less than 3½ inches (88.9 mm) above the roof surface, identified in accordance with Section 401.5, and shall not be concealed. Piping installed above ground, outdoors, and installed across the surface of roofs shall be securely supported to the structure and located where it will be protected from physical damage. Where passing through an outside wall, the piping shall be protected against corrosion in accordance with Section 404.11. Where piping is encased in a protective pipe sleeve, the annular space between the piping and the sleeve shall be sealed. At least one end shall have a vented outlet piped to a safe location outdoors. The vent terminal shall be outdoors, minimum 18 inches (457.2 mm) above grade, not under an opening to building or overhang, and shall be installed so as to prevent the entrance of water and insects.

404.10 Reserved.

404.11 Protection against corrosion. Metallic pipe, fittings, and joints susceptible to corrosive action, such as soil condition or moisture, shall be protected in an approved manner as set forth in Sections 404.11.1 through 404.11.6.

404.11.1 Prohibited use. Threaded or uncoated welded joints shall not be used in piping in contact with soil. Uncoated piping, fittings, and joints shall be prohibited where internal or external corrosion is known to occur.

404.11.2 Protection of exposed pipe. Ferrous metal exposed in exterior locations shall be protected from corrosion by one of the following methods:

1. Zinc coatings (galvanizing).
2. Bonded coatings in accordance with Section 404.11.4.
3. Protective coatings in accordance with Section 404.11.5.
4. Protective wrappings in accordance with Section 404.11.6.

404.11.3 Protection of buried pipe. Ferrous metal installed in buried locations shall be protected from corrosion by one of the following methods:

1. Bonded coatings in accordance with Section 404.11.4.
2. Protective wrappings in accordance with Section 404.11.6.

Where dissimilar metals are joined underground, an insulating coupling or fitting shall be used. Piping shall not be laid in contact with cinders.

404.11.4. Protective bonded coatings. Pipe protective bonded coatings shall be approved for the application and shall be factory applied.

Exception: Where installed in accordance with the manufacturer's instructions, field application of bonded coatings shall be permitted for pipe, fittings and locations where the factory coating or wrapping has been damaged or necessarily removed at joints.

404.11.5 Protective coatings. Pipe protective coatings shall be approved for the application and shall be factory applied or field applied in accordance with the manufacturer's instructions.

Exception: Where installed in accordance with the manufacturer's instructions, field application of coatings shall be permitted for pipe, fittings and locations where the factory coating has been damaged or necessarily removed at joints.

404.11.6 Protective wrappings. Pipe protective wrappings shall be approved for the application and shall be field applied in accordance with the manufacturer's instructions.

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404.12 Minimum burial depth. Underground piping systems shall be installed a minimum depth of 24 inches (609.6 mm) below grade, measured to the top of the pipe.

Exception: Burial depth less than 24 inches (609.6 mm) is permissible when the underground piping system is protected by steel protective plates not less than $\frac{1}{4}$ -inch thick (6.36 mm) and with joints tack welded and overlapped a minimum of 3 inches (76.2 mm). The pipe shall be not less than 4 inches (101.6 mm) below the protective steel plate.

404.13 Trenches. The trench shall be graded so that the pipe has a firm, substantially continuous bearing on the bottom of the trench.

404.13.1 Backfilling. Material used under and beside pipes shall be clean backfill, free of discarded construction material and debris. Loose earth free from rocks, broken concrete and frozen chunks shall be placed in the trench in 6-inch (152.4 mm) layers and tamped in place until the crown of the pipe is covered by 12 inches (304.8 mm) of tamped earth. The backfill under and beside the pipe shall be compacted for pipe support. Backfill shall be brought up evenly on both sides of the pipe so that the pipe remains aligned. In instances where the manufacturer's installation instructions for materials are more restrictive than those prescribed by the code, the material shall be installed in accordance with the more restrictive requirement. Backfilling is subject to progress inspection in accordance with Section 108.

404.14 Piping underground beneath buildings. Piping installed underground beneath buildings is prohibited except where the piping is encased in a conduit of wrought iron or steel pipe designed to withstand the superimposed loads. The conduit shall be protected from corrosion in accordance with Section 404.11 and shall be installed in accordance with Section 404.14.1, 404.14.2 or 404.14.3.

404.14.1 Conduit with one end terminating outdoors. The conduit shall extend into an occupiable portion of the building and, at the point where the conduit terminates in the building, the space between the conduit and the gas piping shall be permanently sealed to prevent the possible entrance of any gas leakage. The conduit shall extend not less than 2 inches (50.8 mm) beyond the point where the pipe emerges from the floor. The end sealing shall be capable of withstanding the full pressure of the gas pipe, and the conduit shall be designed for the same pressure as the pipe. Such conduit shall extend not less than 4 inches (101.6 mm) outside of the building, shall be vented above grade to the outdoors and shall be installed so as to prevent the entrance of water and insects.

404.14.2 Conduit with both ends terminating indoors. Where the conduit originates and terminates within the same building, the conduit shall originate and terminate in an accessible portion of the building and shall not be sealed. The conduit shall extend not less than 2 inches (50.8 mm) beyond the point where the pipe emerges from the floor.

404.14.3 Conduit with both ends terminating indoors within different buildings. Where the conduit originates indoors in a building and terminates indoors within a different building, the conduit shall comply with Section 404.14.2.

404.15 Outlet closures. Gas outlets shall be permitted only under the following conditions:

1. Valved and capped gas tight outlets for appliance outlets listed on approved applications.
2. Valved and capped outlets on each floor in nonproduction laboratory buildings for future laboratories.
3. Listed and labeled flush-mounted-type quick disconnect devices and listed and labeled gas convenience outlets installed in accordance with the manufacturer's instructions.

Exceptions:

1. Test ports for gas riser valves installed downstream of a riser control valve.
2. Outlets for pressure sensors and gauges installed in conjunction with gas appliances or equipment installed in accordance with the manufacturer's instructions.

404.16 Location of outlets. The unthreaded portion of piping outlets shall extend not less than 1 inch (25.4 mm) through finished ceilings and walls and where extending through floors or outdoor patios and slabs, shall be not less than 2 inches (50.8 mm) above them. The outlet fitting or piping shall be securely supported. Outlets shall not be placed behind doors. Outlets shall be located in the room or space where the appliance is installed.

Exception: Listed and labeled flush-mounted-type quick-disconnect devices and listed and labeled gas convenience outlets shall be installed in accordance with the manufacturer's instructions.

404.17 Reserved.

404.18 Pipe cleaning. The use of a flammable or combustible gas to clean or remove debris from the interior of a piping system shall be prohibited.

404.19 Prohibited devices. A device shall not be placed inside the piping or fittings that will reduce the cross-sectional area or otherwise obstruct the free flow of gas.

Exceptions:

1. Approved gas filters.
2. An approved fitting or device where the gas piping system has been sized to accommodate the pressure drop of the fitting or device.
3. Approved gas meters for monitoring and analysis of gas usage.

404.20 Testing of piping. Before any system of piping is put in service, it shall be tested to ensure that it is gas tight. Testing, inspection and purging of piping systems shall comply with Section 406.

404.21 Qualifications for gas work. Gas work shall be performed by individuals possessing the qualifications set forth in Article 28-423 of the *Administrative Code*. Welders must be qualified in accordance with Section 404.21.1.

404.21.1 Welder's qualifications. Welders installing gas piping within buildings at any pressure shall comply with the following:

1. Welders shall be qualified for all pipe sizes, wall thicknesses and all positions in accordance with the ASME *Boiler and Pressure Vessel Code*, Section IX. Requalification of welders is required on an annual basis and when requested by the commissioner.
2. Welder qualification testing shall be performed by an approved agency and the inspector witnessing the test shall be an authorized AWS Certified Welding Inspector. Radiographic test specimens shall be evaluated by a radiographic inspector having a minimum radiography qualification of Level II in accordance with the ASNT, Document No. SNT-TC-1A, Supplement A.
3. Copies of the certified welder qualification reports shall be maintained by both the approved agency and the licensed master plumber employing the welder(s) for at least six years and shall be made available to the department upon request.
4. The approved agency shall submit certified welder qualification reports to the department upon successful qualification of a welder and when requested by the commissioner.
5. The licensed master plumber employing the welder(s) shall submit a statement to the department including who welded the gas piping along with a copy(s) of the certified welder qualification report(s) witnessed by a representative of the licensed master plumber, at the time of the first roughing inspection.
6. Requalification of a welder is required should the welder fail to maintain welder's continuity every six months. The licensed master plumber employing the welder shall maintain a welder continuity log and the log shall be made available to the department upon request.

404.21.2 Welding requirements. All gas distribution piping to customer appliances operating in excess of $\frac{1}{2}$ psig (3.5 kPa gauge) shall be welded.

404.21.3 Welding records. The licensed master plumber employing the welder(s) shall assign to each welder an identification symbol or number to identify the welds performed by that particular welder. The welder shall identify all welds with his or her symbol or number. The licensed master plumber shall maintain records identifying the weld(s) made by each welder for at least six years and shall make such records available to the department upon request.

§§§ **SECTION FGC 405**
PIPING BENDS AND CHANGES IN DIRECTION

405.1 General. Changes in direction of pipe shall be permitted to be made by the use of fittings, factory bends, or field bends.

405.2 Metallic pipe. Metallic pipe bends shall comply with the following:

1. Bends shall be made only with bending tools and procedures intended for that purpose.
2. All bends shall be smooth and free from buckling, cracks or other evidence of mechanical damage.
3. Bends shall be made only with seamless pipe.
4. Pipe shall not be bent through an arc of more than 90 degrees (1.6 rad).
5. The inside radius of a bend shall be not less than six times the outside diameter of the pipe.

405.3 Reserved.

405.4 Elbows. Factory-made welding elbows or transverse segments cut therefrom shall have an arc length measured along the crotch of not less than 1 inch (25.4 mm) in pipe sizes 2 inches (50.8 mm) and larger.

SECTION FGC 406
INSPECTION, TESTING AND PURGING

406.1 General. Prior to acceptance and initial operation, all piping installations shall be inspected and pressure tested to determine that the materials, design, fabrication and installation practices comply with the requirements of this code.

406.1.1 Inspections. Inspection shall consist of visual examination during or after manufacture, fabrication, assembly or pressure tests as appropriate. Supplementary types of nondestructive inspection techniques, such as magnetic-particle, radiographic, ultrasonic, etc., shall not be required unless specifically listed herein or in the engineering design. All inspections shall be performed in accordance with Section 108.

406.1.1.1 Welding inspections and testing. Welded gas piping shall be subject to special inspection in accordance with Chapter 17 of the *New York City Building Code*. Radiographic testing shall be performed on all butt welds in gas meter and gas distribution piping operating at pressures exceeding 5 psig (34.5 kPa gauge) within buildings, in accordance with *ASME Boiler and Pressure Vessel Code*, Section IX.

406.1.2 Repairs and additions. In the event repairs or additions are made after the pressure test, the affected piping shall be tested. All sections of de-energized piping shall be pressure tested in accordance with Section 406.4.

406.1.3 New branches. Where new branches are installed, the new branches shall be pressure tested. Connections between the new piping and the existing piping shall be pressure tested.

406.1.4 Section testing. A piping system shall be permitted to be tested as a complete unit or in sections. Under no circumstances shall a valve in a line be used as a bulkhead between gas in one section of the piping system and test medium in an adjacent section, except where a double block and bleed valve system or flange is installed. A valve shall not be subjected to the test pressure unless the piping system upstream of the valve has been deenergized. The section of piping not subjected to the pressure test shall be checked with a noncorrosive leak-detecting fluid or other approved leak-detecting methods. Sections may be tested at different pressures in accordance with Section 406.4.

406.1.5 Regulators and valve assemblies. Regulator and valve assemblies fabricated independently of the piping system in which they are to be installed shall be permitted to be tested with inert gas or air at the time of fabrication. Such regulators and valve assemblies shall not be subjected to the testing requirements of Section 406.4. Spool pieces shall be installed in place for the test. The regulators and valve assemblies not subjected to the pressure test shall be checked with a noncorrosive leak-detecting fluid or other approved leak-detecting methods.

406.1.6 Pipe clearing. Prior to testing, the interior of the pipe shall be purged with air or an inert gas to clear foreign material, including weld splatter, dirt, rags, and other debris left inside the pipe during welding operations or piping installation.

406.2 Test medium. The test medium shall be air, nitrogen, carbon dioxide or an inert gas. Oxygen shall not be used.

406.3 Test preparation. Pipe joints and piping including welds, shall be left exposed for examination during the test.

Exception: Covered or concealed pipe end joints that have been previously inspected in accordance with this code.

406.3.1 Expansion joints. Expansion joints shall be provided with temporary restraints, if required, for the additional thrust load under test.

406.3.2 Appliance and equipment isolation. Appliances and equipment that are not to be included in the test shall be either disconnected from the piping or isolated by blanks, blind flanges or caps. Flanged joints at which blinds are inserted to blank off other equipment during the test shall not be required to be tested. The affected piping and joints shall be checked with a noncorrosive leak-detecting fluid or other approved leak-detecting methods when gas is energized.

406.3.3 Appliance and equipment disconnection. Where the piping system is connected to appliances or equipment designed for operating pressures of less than the test pressure, such appliances or equipment shall be isolated from the piping system by disconnecting them and capping the outlet(s). The affected piping and joints shall be checked with a noncorrosive leak-detecting fluid or other approved leak-detecting methods when gas is energized.

406.3.4 Valve isolation. Where the piping system is connected to appliances or equipment designed for operating pressures equal to or greater than the test pressure, such appliances or equipment shall be isolated from the piping system by closing the individual appliance or equipment shutoff valve(s). The affected piping and joints shall be checked with a noncorrosive leak-detecting fluid or other approved leak-detecting methods when gas is energized.

406.3.5 Testing precautions. Testing of piping systems shall be performed in a manner that protects the safety of employees and the public during the test. Bulkheads, anchorage, and bracing suitably designed to resist test pressures shall be installed if necessary. Prior to testing, the interior of the pipe shall be purged to flush out all foreign material, including weld splatter, dirt, rags, and other debris left inside the pipe during welding operations and piping installation.

406.4 Test pressure measurement. Upon completion of the installation of a section of a gas system or of the entire gas system, and before appliances are connected thereto, the completed section or system shall be verified as to materials, and tested and proven tight. Test pressure shall be measured with a manometer or with a pressure-measuring device designed and calibrated to read, record or indicate a pressure loss caused by leakage during the pressure test period. The source of pressure shall be isolated before the pressure tests are made. Mechanical gauges used to measure test pressures shall have a range such that the highest end of the scale is not greater than two times the test pressure. Gauges shall comply with the requirements of Section 406.4.3 or 406.4.4.

406.4.1 Test pressure. The test pressure to be used shall be not less than 1½ times the proposed maximum working pressure, but not less than 3 psig (20.9 kPa gauge), irrespective of design pressure. Where the test pressure exceeds 125 psig (861.8 kPa gauge), the test pressure shall not exceed a value that produces a hoop stress in the piping greater than 50 percent of the specified minimum yield strength of the pipe.

Exceptions:

1. Gas piping with distribution pressures above 5 psig (34.5 kPa gauge) shall be tested to twice the maximum allowable operating pressure, but not less than 100 psig (689.5 kPa gauge), for a minimum of 1 hour.
2. Gas piping installed in public corridors in accordance with Section 404 shall be tested and proven tight at a pressure of 10 psig (68.9 kPa gauge) for a minimum of 30 minutes.
3. Meter piping shall be pressure tested in accordance with the requirements of the serving utility. These requirements shall be either the same as those for testing distribution piping in this section or, if different, the piping shall be certified by the utility company as being tested in compliance with their requirements.
4. Notwithstanding the above, all field applied or factory applied coated and wrapped pipe shall be pressure tested at a minimum of 100 psig (689.5 kPa gauge) for a minimum of 60 minutes. For testing, the piping

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shall be filled with air or an inert gas, and the source of pressure shall be isolated before the pressure readings are made. All test duration time periods are to be measured after stabilization of testing medium.

406.4.1.1 Factory applied coatings. Gas piping with factory applied coatings intended to protect against corrosion during material shipping and storage shall be tested at not less than $1\frac{1}{2}$ times the proposed maximum working pressure, but not less than 3 psig (20.9 kPa gauge), irrespective of design pressure. Where the test pressure exceeds 125 psig (861.8 kPa gauge), the test pressure shall not exceed a value that produces a hoop stress in the piping greater than 50 percent of the specified minimum yield strength of the pipe.

406.4.1.2 Field applied paint. Gas piping with field applied paint intended to protect against corrosion shall be tested at not less than $1\frac{1}{2}$ times the proposed maximum working pressure, but not less than 3 psig (20.9 kPa gauge), irrespective of design pressure. Where the test pressure exceeds 125 psig (861.8 kPa gauge), the test pressure shall not exceed a value that produces a hoop stress in the piping greater than 50 percent of the specified minimum yield strength of the pipe.

406.4.2 Test duration. Test duration shall be not less than $\frac{1}{2}$ hour, unless a greater duration is required by Section 406.4.1.

406.4.3 Minimum standards for gauges. This section establishes minimum standards for gauges used to test gas piping. Each gauge shall meet the following requirements:

1. The gauge shall be manufactured and used in accordance with ASME B40.100, which incorporates ASME B40.1 and ASME B40.7, and the manufacturer shall provide with the gauge a written statement that the gauge is manufactured in accordance with such ASME standard;
2. The gauge shall be labeled with the name of the manufacturer;
3. The units of measurement “psig” shall appear on the face of the gauge; and
4. The gauge shall be kept in good working order.

406.4.4 Digital gauges used to measure pressure in the magnitude of 3 psig (20.9 kPa gauge) and higher. Each digital gauge used to measure pressure in the magnitude of 3 psig (20.9 kPa gauge) and higher shall meet the following requirements in addition to satisfying the minimum requirements set forth in Section 406.4.3:

1. The gauge shall have a minimum reading of $\frac{1}{100}$ of a psig (69 Pa), and
2. An extra charged battery shall be readily available for immediate use with the gauge.

†††**406.4.5 Witnessing tests of gas piping systems.** Tests of gas piping systems in accordance with this code shall be witnessed by department plumbing inspectors.

†††**406.4.6 Conducting tests of gas piping systems.** Tests of gas piping systems in accordance with this code shall be conducted by an individual with not less than five years’ experience in gas work.

406.4.7 Notification. The holder of the plumbing permit shall give at least two days prior notice to the commissioner that the plumbing work covered by the permit is ready for inspections and test.

406.5 Detection of leaks and defects. The piping system shall withstand the test pressure specified without showing any evidence of leakage or other defects. Any reduction of test pressures as indicated by pressure gauges shall be deemed to indicate the presence of a leak unless such reduction can be readily attributed to some other cause.

406.5.1 Detection methods. The leakage shall be located by means of an approved gas detector, a noncorrosive leak detection fluid or other approved leak detection methods.

406.5.2 Corrections. Where leakage or other defects are located, the affected portion of the piping system shall be repaired or replaced and retested.

406.6 Piping system and equipment leakage check. Leakage checking of systems and equipment shall be in accordance with Sections 406.6.1 through 406.6.4.

406.6.1 Check gases. Leak checks using fuel gas shall be permitted in piping systems that have been pressure tested in accordance with Section 406.

406.6.2 Before turning gas on. During the process of turning gas on into a system of new gas piping, the entire system shall be inspected to determine that there are no open fittings or ends and that all valves at unused outlets are closed and plugged or capped.

406.6.2.1 Gas authorization required. A gas piping system shall not be energized prior to authorization. Such authorization is demonstrated by issuance of a certificate of approval of gas installation by the department. The certificate shall not be issued prior to the completion of the required inspections and tests. The certificate shall not expire.

Exception: Gas authorization shall not be required when gas is reestablished by a utility company as permitted under Section 28-105.4.3 of Chapter 1 of Title 28 of the *Administrative Code*.

406.6.2.1.1 Existing gas piping. Any gas piping system or part thereof that has been previously energized and has been shut down shall be required to obtain a certificate of approval of gas installation for all appliances served by the affected gas meter(s) prior to the reestablishment of gas into such system.

Exception: When an existing gas appliance is replaced and no gas piping is replaced upstream of the existing appliance shutoff valve and not more than 6 feet (1828.8 mm) of pipe is installed or replaced downstream of the existing appliance shutoff valve, a certificate of approval of gas installation shall not be required to energize such pipe or replacement appliance.

406.6.2.1.2 Partial gas authorization. A certificate of approval of gas installation may be issued for a portion of a gas piping system provided the remainder of the system is locked-off by the gas service provider.

406.6.2.2 Establishing gas supply. It shall be unlawful for any utility company to supply or restore gas to a building, place or premises in which new meters other than replacement gas meters are required until a certificate of approval of gas installation is issued in accordance with Section 406.6.2.1. When new gas service piping has been installed, it shall be locked-off by the utility either by locking the gas service line valve or by installing a locking device on the outside gas service line valve. The lock shall not be removed until the gas meter piping (other than utility-owned) and gas distribution piping has been inspected and certified as required by the department as being ready for service. When an existing gas service piping has been taken out of service, it shall be locked-off by the utility. The lock shall not be removed until the gas meter piping (other than utility-owned) and gas distribution piping has been inspected and certified as required by the department as being ready for service.

406.6.2.3 Alterations to gas piping systems. When alterations, extensions or repairs to existing gas meter piping or gas distribution piping requires the operation of a service head valve, regulator valve or meter valve that would interrupt the flow of gas to a building, the utility shall be notified by the owner or his or her authorized representative.

406.6.2.4 Emergency service head valve, regulator valve or gas meter valve operation. It is permissible for a licensed plumber or his or her employee to operate a valve in an emergency to alleviate a hazardous condition. The utility shall be notified by the licensee or his or her authorized representative. Such licensee or authorized representative shall wait on location for utility company operators to arrive and lock the affected valve.

406.6.3 Leak check. Immediately after the gas is turned on into a new system or into a system that has been initially restored after an interruption of service, the piping system shall be checked for abnormal operating conditions and leakage. The leakage check shall be performed by means of a combustible gas indicator complying with Section 406.7.1.4, or a noncorrosive leak detection fluid. Where leakage is indicated, or where an abnormal operating condition presents a hazard, the gas supply shall be shut off until the necessary repairs have been made.

406.6.4 Placing appliances and equipment in operation. Appliances and equipment shall not be placed in operation until after the piping system has been checked in accordance with Section 406.6.3, the piping system has been purged in accordance with Section 406.7 and the connections to the appliances have been checked for leakage.

406.6.4.1 Requirements for placing appliances in operation. The following will be required prior to placing appliances in operation as applicable:

1. Required fire protection system (sprinkler or standpipe) are completed, inspected and ready for service.
2. Such appliance and related equipment and gas piping are inspected by the department.
3. Associated fire suppression system is inspected and approved by the Fire Department.

Exception: Appliances may be used as temporary sources of heat during construction as needed and in accordance with requirements of the department.

GAS PIPING INSTALLATIONS

406.7 Purging. The purging of piping shall be in accordance with Sections 406.7.1 through 406.7.3.

406.7.1 Piping systems required to be purged outdoors. The purging of piping systems shall be in accordance with the provisions of Sections 406.7.1.1 through 406.7.1.4 where the piping system meets either of the following:

1. The design operating gas pressure is greater than 2 psig (13.8 kPa).
2. The piping being purged contains one or more sections of pipe meeting the size and length criteria of Table 406.7.1.1.

406.7.1.1 Removal from service. Where existing gas piping is opened, the section that is opened shall be isolated from the gas supply and the line pressure vented in accordance with Section 406.7.1.3. Where gas piping meeting the criteria of Table 406.7.1.1 is removed from service, the residual fuel gas in the piping shall be displaced with an inert gas.

**TABLE 406.7.1.1
SIZE AND LENGTH OF PIPING**

NOMINAL PIPE SIZE (inches)	LENGTH OF PIPING (feet)
$\geq 2\frac{1}{2} < 3$	≥ 50
$\geq 3 < 4$	≥ 30
$\geq 4 < 6$	≥ 15
$\geq 6 < 8$	≥ 10
≥ 8	Any length

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

406.7.1.2 Placing in operation. Where gas piping containing air and meeting the criteria of Table 406.7.1.1 is placed in operation, the air in the piping shall first be displaced with an inert gas. The inert gas shall then be displaced with fuel gas in accordance with Section 406.7.1.3.

406.7.1.3 Outdoor discharge of purged gases. The open end of a piping system being pressure vented or purged shall discharge directly to an outdoor location. Purging operations shall comply with all of the following requirements:

1. The point of discharge shall be controlled with a shutoff valve.
2. The point of discharge shall be located not less than 10 feet (3048 mm) from sources of ignition, not less than 10 feet (3048 mm) from building openings and not less than 25 feet (7620 mm) from mechanical air intake openings.
3. During discharge, the open point of discharge shall be continuously attended and monitored with a combustible gas indicator that complies with Section 406.7.1.4.
4. Purging operations introducing fuel gas shall be stopped when at least 95 percent fuel gas by volume is detected within the pipe.
5. Persons not involved in the purging operations shall be evacuated from all areas within 10 feet (3048 mm) of the point of discharge.

406.7.1.4 Combustible gas indicator. Combustible gas indicators shall be listed and shall be calibrated in accordance with the manufacturer's instructions. Combustible gas indicators shall numerically display a volume scale from zero percent to 100 percent in 1 percent or smaller increments.

406.7.2 Piping systems allowed to be purged indoors or outdoors. The purging of piping systems shall be in accordance with the provisions of Section 406.7.2.1 where the piping system meets both of the following:

1. The design operating gas pressure is 2 psig (13.79 kPa) or less.
2. The piping being purged is constructed entirely from pipe or tubing not meeting the size and length criteria of Table 406.7.1.1.

406.7.2.1 Purging procedure. The piping system shall be purged in accordance with one or more of the following:

1. The piping shall be purged with fuel gas that is discharged to the indoors or outdoors, and the point of discharge shall be monitored with a listed combustible gas detector in accordance with Section 406.7.2.2. Purging shall be stopped when fuel gas is detected.
2. The piping shall be purged in accordance with written procedures of the utility company.

406.7.2.2 Combustible gas detector. Combustible gas detectors shall be listed and shall be calibrated or tested in accordance with the manufacturer's instructions. Combustible gas detectors shall be capable of indicating the presence of fuel gas.

406.7.3 Purging appliances and equipment. After the piping system has been placed in operation, appliances and equipment subsequently installed shall be purged before being placed into operation.

SECTION FGC 407 PIPING SUPPORT

407.1 General. Piping shall be provided with support in accordance with Section 407.2. In addition, when earthquake loads are applicable in accordance with the *New York City Building Code*, a detailed piping system stress analysis including seismic analysis shall be performed. The pipe supports and restraints shall be designed and installed to accommodate the resultant seismic forces, moments and displacements from this stress analysis in accordance with the *New York City Building Code*.

407.2 Design and installation. Piping shall be supported with metal pipe hooks, metal pipe straps, metal bands, metal brackets, metal hangers or building structural components, suitable for the size of piping, of adequate strength and quality, and located at intervals so as to prevent or damp out excessive vibration. Piping shall be anchored to prevent undue strains on connected appliances and shall not be supported by other piping. Pipe hangers and supports shall conform to the requirements of MSS SP-58 and shall be spaced in accordance with Section 415. Supports, hangers and anchors shall be installed so as not to interfere with the free expansion and contraction of the piping between anchors. All parts of the supporting equipment shall be designed and installed so that they will not be disengaged by movement of the supported piping.

SECTION FGC 408 DRIPS AND SLOPED PIPING

408.1 Slopes. Piping for other than dry gas conditions shall be sloped not less than $\frac{1}{4}$ inch in 15 feet (6.3 mm in 4572 mm) to prevent traps. The local gas supplier/utility company should be consulted to determine the type of fuel gas available for the intended service.

408.2 Drips. Drips are not permitted.

408.3 Test fittings. A manufactured test fitting shall be installed where required by the gas supplier. Such fitting shall be downstream of a lockable supply or riser valve in accordance with the requirements of the gas supplier and the manufacturer's instructions. The outlet of the test fitting shall be the smallest commercially available. The installation of manufactured test fittings in other locations shall be prohibited.

408.4 Sediment trap. Where a sediment trap is not incorporated as part of the appliance, a sediment trap shall be installed downstream of the appliance shutoff valve as close to the inlet of the appliance as practical. The sediment trap shall be either a tee fitting having a capped nipple of any length installed vertically in the bottom most opening of the tee as illustrated in Figure 408.4 or other device approved as an effective sediment trap. Illuminating appliances, ranges, clothes dryers, decorative vented appliances for installation in vented fireplaces, gas fireplaces and outdoor grills need not be so equipped.

GAS PIPING INSTALLATIONS

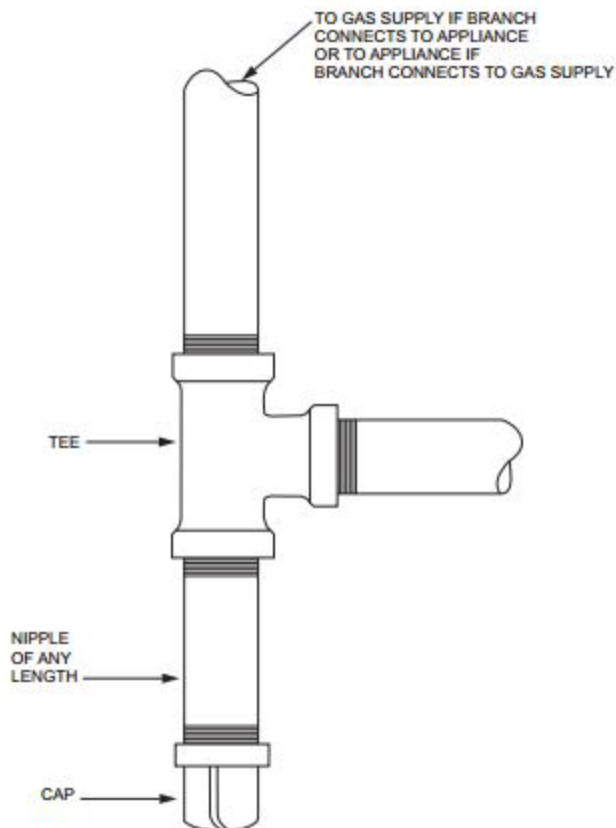


FIGURE 408.4
METHOD OF INSTALLING A TEE FITTING SEDIMENT TRAP

SECTION FGC 409 **SHUTOFF VALVES**

409.1 General. Piping systems shall be provided with shutoff valves in accordance with this section.

409.1.1 Valve approval. Shutoff valves shall be of an approved type; shall be constructed of materials compatible with the piping; and shall comply with the standard that is applicable for the pressure and application, in accordance with Table 409.1.1.

TABLE 409.1.1
MANUAL GAS VALVE STANDARDS

VALVE STANDARDS	APPLIANCE SHUTOFF VALVE APPLICATION UP TO 1/2 psig PRESSURE	OTHER VALVE APPLICATIONS			
		UP TO 1/2 psig PRESSURE	UP TO 2 psig PRESSURE	UP TO 5 psig PRESSURE	UP TO 125 psig PRESSURE
ANSI Z21.15/CGA 9.1	X	—	—	—	—
ASME B16.44	X	X	X ^a	X ^b	—
ASME B16.33	X	X	X	X	X

For SI: 1 pound per square inch gauge = 6.895 kPa.

a. If labeled 2G.

b. If labeled 5G.

409.1.2 Prohibited locations. Shutoff valves shall be prohibited in concealed locations and furnace plenums.

409.1.3 Access to shutoff valves. Shutoff valves shall be located in places so as to provide access for operation and shall be installed so as to be protected from damage.

409.2 Meter valve. Every meter shall be equipped with a shutoff valve located on the supply side of the meter. Meter valves shall be of a type approved by the utility company.

409.3 Shutoff valves for multiple-house line systems. Where a single meter is used to supply gas to more than one building or tenant, a separate shutoff valve shall be provided for each building or tenant.

409.3.1 Multiple tenant buildings. In multiple tenant buildings, where a common piping system is installed to supply other than individual dwelling units, shutoff valves shall be provided for each tenant. Each tenant shall have access to the shutoff valve serving that tenant's space.

409.3.2 Individual buildings. In a common system serving more than one building, shutoff valves shall be installed outdoors at each building.

409.3.3 Identification of shutoff valves. Each house line shutoff valve shall be plainly marked with an identification tag attached by the installer so that the piping systems supplied by such valves are readily identified. Identification shall be provided in compliance with ASME A13.1.

409.4 MP regulator valves. An approved shutoff valve shall be installed immediately ahead of each MP regulator.

409.5 Appliance shutoff valve. Each appliance shall be provided with a shutoff valve in accordance with Section 409.5.1 or 409.5.2.

409.5.1 Located within the same room. The shutoff valve shall be located in the same room as the appliance. The shutoff valve shall be within 6 feet (1828.8 mm) of the appliance, and shall be installed upstream of the union, connector or quick disconnect device it serves. Such shutoff valves shall be provided with access and located not more than 60 inches (1524 mm) above the finished floor. Appliance shutoff valves located in the firebox of a fireplace shall be installed in accordance with the appliance manufacturer's instructions. Where the shutoff valve is located in the firebox, a remote shutoff valve shall also be provided.

Exception: Shutoff valves may be located more than 60 inches (1524 mm) above the finished floor where the gas connection to the appliance is more than 60 inches (1524 mm) above the finished floor.

409.5.2 Vented decorative appliances and room heaters. Shutoff valves for vented decorative appliances, room heaters and decorative appliances for installation in vented fireplaces shall be permitted to be installed in an area remote from the appliances where such valves are provided with ready access. Such valves shall be permanently identified and shall not serve another appliance. The piping from the shutoff valve to within 6 feet (1828.8 mm) of the appliance shall be designed, sized and installed in accordance with Sections 401 through 408.

409.6 Shutoff valve for laboratories. Where provided with two or more fuel gas outlets, including table-, bench- and hood-mounted outlets, each laboratory space in educational, research, commercial and industrial occupancies shall be provided with a single dedicated shutoff valve through which all such gas outlets shall be supplied. The dedicated shutoff valve shall be readily accessible, located within the laboratory space served, located adjacent to the egress door from the space and shall be identified by approved signage stating "Gas Shutoff."

409.7 Outside gas cut-off. An outside gas service line valve or other outside shutoff device or method acceptable to the commissioner shall be provided in accordance with Section E.2 of Appendix E.

409.8 Riser shutoff valves. Risers shall be equipped with an approved lockable shutoff valve. There shall be a test port installed downstream of the valve.

409.9 Branch shutoff valves. Branches serving more than one appliance shall be equipped with an approved lockable shutoff valve. There shall be a test port installed downstream of the valve.

SECTION FGC 410 FLOW CONTROLS

GAS PIPING INSTALLATIONS

410.1 Pressure regulators. A line pressure regulator shall be installed where the appliance is designed to operate at a lower pressure than the supply pressure. Line gas pressure regulators shall be listed as complying with ANSI Z21.80/CSA 6.22. Access shall be provided to pressure regulators. Pressure regulators shall be protected from physical damage. Regulators installed on the exterior of the building shall be approved for outdoor installation.

410.2 MP regulators. Medium pressure (MP) regulators shall comply with the following:

1. The MP regulator shall be approved and shall be suitable for the inlet and outlet gas pressures for the application.
2. The MP regulator shall maintain a reduced outlet pressure under lockup (no-flow) conditions.
3. The capacity of the MP regulator, determined by published ratings of its manufacturer, shall be adequate to supply the appliances served.
4. The MP regulator shall be provided with access. Where located indoors, the regulator shall be vented to the outdoors or shall be equipped with a leak-limiting device, in either case complying with Section 410.3.
5. A tee fitting with one opening capped or plugged shall be installed between the MP regulator and its upstream shutoff valve. Such tee fitting shall be positioned to allow connection of a pressure-measuring instrument and to serve as a sediment trap.
6. A tee fitting with one opening capped or plugged shall be installed not less than 10 pipe diameters downstream of the MP regulator outlet. Such tee fitting shall be positioned to allow connection of a pressure-measuring instrument. A pressure gauge with an isolation valve may also be installed in addition to the fitting.
7. At least one union shall be installed within 1 foot (304.8 mm) of either side of the MP regulator.

410.3 Venting of regulators. Pressure regulators that require a vent shall be vented directly to the outdoors. The vent terminal shall be a minimum of 18 inches (457.2 mm) above grade, not under a building-opening or overhang, and shall be installed so as to prevent the entrance of water and insects.

Exception: A vent to the outdoors is not required for regulators equipped with and labeled for utilization with an approved vent-limiting device installed in accordance with ANSI Z21.18 and the manufacturer's instructions. The vent-limiting device shall be installed in a location sufficiently ventilated to dissipate any gas discharged from the device.

410.3.1 Vent piping. Vent piping for relief vents and breather vents shall be constructed of materials allowed for gas piping in accordance with Section 403. Vent piping shall be not smaller than the vent connection on the pressure-regulating device. Vent piping serving relief vents and combination relief and breather vents shall be run independently to the outdoors and shall serve only a single device vent. Vent piping serving only breather vents is permitted to be connected in a manifold arrangement where sized in accordance with an approved design that minimizes back pressure in the event of diaphragm rupture. Regulator vent piping shall not exceed the length specified in the regulator manufacturer's instructions.

410.4 Excess flow valves. Where automatic excess flow valves are installed, they shall be listed for the application and shall be sized and installed in accordance with the manufacturer's instructions.

410.5 Flashback arrestor check valve. Where fuel gas is used with oxygen in any hot work operation, a listed protective device that serves as a combination flashback arrestor and backflow check valve shall be installed at an approved location on both the fuel gas and oxygen supply lines. Where the pressure of the piped fuel gas supply is insufficient to ensure such safe operation, approved equipment shall be installed between the gas meter and the appliance that increases pressure to the level required for such safe operation.

SECTION FGC 411 APPLIANCE AND MANUFACTURED HOME CONNECTIONS

411.1 Connecting appliances. Except as required by Section 411.1.1, appliances shall be connected to the piping system by one of the following:

1. Rigid metallic pipe and fittings. Space heaters and water heaters shall be connected using rigid piping and fittings only. Use of semirigid (flexible) metallic tubing, fittings, appliance connectors or quick-disconnect devices is not permitted for this application.
2. Listed and labeled appliance connectors in compliance with ANSI Z21.24/CGA 6.10 and installed in accordance with the manufacturer's instructions and located entirely in the same room as the appliance. Connectors shall have an overall length not to exceed 6 feet (1828.8 mm). Measurement shall be made along the centerline of the connector. Only one connector shall be used for each appliance.
3. Listed and labeled quick-disconnect devices used in conjunction with listed and labeled appliance connectors.
4. Listed and labeled convenience outlets used in conjunction with listed and labeled appliance connectors.
5. Listed and labeled outdoor appliance connectors in compliance with ANSI Z21.75/CSA 6.27 and installed in accordance with the manufacturer's instructions.
6. Listed outdoor gas hose connectors in compliance with ANSI Z21.54 used to connect portable outdoor appliances. The gas hose connection shall be made only in the outdoor area where the appliance is used, and shall be to the gas piping supply at an appliance shutoff valve, a listed quick-disconnect device or listed gas convenience outlet.

411.1.1 Commercial cooking appliances. Commercial cooking appliances installed on casters and appliances that are moved for cleaning and sanitation purposes shall be connected to the piping system with an appliance connector listed as complying with ANSI Z21.69/CSA 6.16 and NSF/ANSI 169 or in accordance with Item 1 of Section 411.1. The commercial cooking appliance connector installation shall be configured in accordance with the manufacturer's instructions. Movement of appliances with casters shall be limited by a restraining device installed in accordance with the connector and appliance manufacturer's instructions.

411.1.2 Protection against damage. Connectors and tubing shall be installed so as to be protected against physical damage.

411.1.3 Connector installation. Fuel connectors shall be installed in accordance with the manufacturer's instructions and Sections 411.1.3.1 through 411.1.3.4.

411.1.3.1 Maximum length. Connectors shall have an overall length not to exceed 6 feet (1828.8 mm). Measurement shall be made along the centerline of the connector. Only one connector shall be used for each appliance.

411.1.3.2 Minimum size. Connectors shall have the capacity for the total demand of the connected appliance.

411.1.3.3 Prohibited locations and penetrations. Connectors shall not be concealed within, or extended through, walls, floors, partitions, ceilings or appliance housings.

Exceptions:

1. Connectors constructed of materials allowed for piping systems in accordance with Section 403 shall be permitted to pass through walls, floors, partitions and ceilings where installed in accordance with Section 409.5.2.
2. Rigid steel pipe connectors shall be permitted to extend through openings in appliance housings.
3. Fireplace inserts that are factory equipped with grommets, sleeves or other means of protection in accordance with the listing of the appliance.
4. Listed connectors shall be permitted to extend through an opening in an appliance housing, cabinet or casing where the tubing or connector is protected against damage.

411.1.3.4 Shutoff valve. A shutoff valve not less than the nominal size of the piping system shall be installed ahead of the connector in accordance with Section 409.5.

411.1.4 Movable appliances. Where appliances are equipped with casters or are otherwise subject to periodic movement or relocation for purposes such as routine cleaning and maintenance, such appliances shall be connected to the supply system piping by means of an appliance connector listed as complying with ANSI Z21.69/CSA 6.16 or by means of Item 1 of Section 411.1 of this code. Such flexible connectors shall be installed and protected against physical damage in accordance with the manufacturer's instructions.

GAS PIPING INSTALLATIONS

411.1.5 Connection of gas engine-powered air conditioners. Internal combustion engines shall not be rigidly connected to the gas supply piping.

411.1.6 Unions. A union fitting shall be provided for appliances connected by rigid metallic pipe. Such unions shall be accessible and located within 6 feet (1828.8 mm) of the appliance.

411.2 Manufactured home connections. Manufactured homes shall be connected to the distribution piping system by one of the following materials:

1. Metallic pipe in accordance with Section 403.4.
2. Listed and labeled connectors in compliance with ANSI Z21.75/CSA 6.27 and installed in accordance with the manufacturer's instructions.

411.3 Suspended low-intensity infrared tube heaters. Suspended low-intensity infrared tube heaters shall be connected to the building piping system with a connector listed for the application complying with ANSI Z21.24/CGA 6.10. The connector shall be installed as specified by the tube heater manufacturer's instructions.

SECTION FGC 412 LIQUEFIED PETROLEUM GAS MOTOR VEHICLE FUEL-DISPENSING FACILITIES

412.1 Applicability. Motor fuel-dispensing facilities for LP-gas fuel shall be designed and operated in accordance with the *New York City Fire Code*.

412.2 Reserved.

412.3 Reserved.

412.4 Reserved.

412.5 Reserved.

412.6 Reserved.

412.7 Reserved.

412.8 Reserved.

SECTION FGC 413 COMPRESSED NATURAL GAS MOTOR VEHICLE FUEL-DISPENSING FACILITIES

413.1 Applicability. Motor fuel-dispensing facilities for CNG fuel shall be designed and operated in accordance with the *New York City Fire Code*.

413.2 Reserved.

413.3 Reserved.

413.4 Reserved.

413.5 Reserved.

413.6 Reserved.

413.7 Reserved.

413.8 Reserved.

SECTION FGC 414 RESERVED

SECTION FGC 415 PIPING SUPPORT INTERVALS

415.1 Interval of support. Piping shall be supported at intervals not exceeding the spacing specified in Table 415.1.

**TABLE 415.1
SUPPORT OF PIPING**

STEEL PIPE, NOMINAL SIZE OF PIPE (inches)	SPACING OF SUPPORTS (feet)
$\frac{1}{2}$	6
$\frac{3}{4}$ or 1	8
$1\frac{1}{4}$ or larger (horizontal)	12
All vertical piping	Every floor level

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

SECTION FGC 416 OVERPRESSURE PROTECTION DEVICES

416.1 Where required. Where the serving gas supplier delivers gas at a pressure greater than 2 psi (13.79 kPa) for piping systems serving appliances designed to operate at a gas pressure of 14 inches w.c. (3.49 kPa) or less, overpressure protection devices shall be installed. Piping systems serving appliances designed to operate at inlet pressures greater than 14 inches w.c. (3.49 kPa) shall be equipped with overpressure protection devices as required by the appliance manufacturer's instructions.

416.2 Pressure limitation requirements. The requirements for pressure limitation shall be in accordance with Sections 416.2.1 through 416.2.5.

416.2.1 Pressure under 14 inches w.c. Where piping systems serving appliances designed to operate with a gas supply pressure of 14 inches w.c. (3.49 kPa) or less are required to be equipped with overpressure protection by Section 416.1, each overpressure protection device shall be adjusted to limit the gas pressure to each connected appliance to 2 psi or less upon a failure of the line pressure regulator.

416.2.2 Pressure over 14 inches w.c. Where piping systems serving appliances designed to operate with a gas supply pressure greater than 14 inches w.c. (3.49 kPa) are required to be equipped with overpressure protection by Section 416.1, each overpressure protection device shall be adjusted to limit the gas pressure to each connected appliance as required by the appliance manufacturer's instructions.

416.2.3 Device capability. Each overpressure protection device installed to meet the requirements of this section shall be capable of limiting the pressure to its connected appliance(s) as required by Section 416.2.1, independently of any other pressure control equipment in the piping system.

416.2.4 Failure detection. Each gas piping system for which an overpressure protection device is required by Section 416 shall be designed and installed so that a failure of the primary pressure control device(s) is detectable.

416.2.5 Relief valve. Where a pressure relief valve is used to meet the requirements of Section 416, it shall have a flow capacity such that the pressure in the protected system is maintained at or below the limits specified in Section 416.2.1 under all of the following conditions:

1. The line pressure regulator for which the relief valve is providing overpressure protection has failed wide open.
2. The gas pressure at the inlet of the line pressure regulator for which the relief valve is providing overpressure protection is not less than the regulator's normal operating inlet pressure.

GAS PIPING INSTALLATIONS

416.3 Devices. Pressure-relieving or pressure-limiting devices shall be one of the following:

1. Pressure relief valve.
2. Monitoring regulator.
3. Series regulator installed upstream from the line regulator and set to continuously limit the pressure on the inlet of the line regulator to the maximum values specified by Section 416.2.1.
4. Automatic shutoff device installed in series with the line pressure regulator and set to shut off when the pressure on the downstream piping system reaches the maximum values specified by Section 416.2.1. This device shall be designed so that it will remain closed until manually reset.

The devices specified in this section shall be installed either as an integral part of the service or line pressure regulator or as separate units. Where separate pressure-relieving or pressure-limiting devices are installed, they shall comply with Sections 416.3.1 through 416.3.6.

416.3.1 Construction and installation. Pressure-relieving and pressure-limiting devices shall be constructed of materials so that the operation of the devices will not be impaired by corrosion of external parts by the atmosphere or of internal parts by the gas. Pressure-relieving and pressure-limiting devices shall be designed and installed so that they can be operated to determine whether the valve is free. The devices shall be designed and installed so that they can be tested to determine the pressure at which they will operate and examined for leakage when in the closed position. The devices shall be maintained and inspected in accordance with Section 416.3.7.

416.3.2 External control piping. External control piping shall be designed and installed so that damage to the control piping of one device will not render both the regulator and the overpressure protection device inoperative.

416.3.3 Setting. Each pressure-relieving or pressure-limiting device shall be set so that the gas pressure supplied to the connected appliances does not exceed the limits specified in Section 416.2.1.

416.3.4 Unauthorized operation. Precautions shall be taken to prevent unauthorized operation of any shutoff valve that will make a pressure-relieving valve or pressure-limiting device inoperative. The valve shall be locked or continuously monitored in the open position.

416.3.5 Vents. The discharge stacks, vents and outlet parts of all pressure-relieving and pressure-limiting devices shall be located so that gas is safely discharged to the outdoors. Discharge stacks and vents shall be designed to prevent the entry of water, insects and other foreign material that could cause blockage. The discharge stack or vent line shall be not less than the same size as the outlet of the pressure-relieving device.

416.3.6 Size of fittings, pipe and openings. The fittings, pipe and openings located between the system to be protected and the pressure-relieving device shall be sized to prevent hammering of the valve and to prevent impairment of relief capacity.

416.3.7 Device maintenance. The overpressure protection device: (i) shall be properly maintained and inspected every two years by a licensed master plumber; (ii) inspection records shall be maintained by the owner and made available to the department; (iii) inspection procedures shall be in accordance with the commissioner; and (iv) replacements or repairs shall be promptly made.

Exception: Where the gas pressure is 15 psig (103.4 kPa gauge) or above, the overpressure protection device shall be inspected annually.